CECS 229 – Assignment 7

Vectors

Given the file, 'pa7.py', containing an incomplete class Vec, complete the code for the following methods:

1. `__abs__(self)` – this method calculates and returns the magnitude of the vector object.
    a. The returned value should be the Euclidian norm of the vector object.
    b. Iterate over the elements of the vector to calculate the Euclidian norm.
2. `__add__(self, other)` – overloads the '+' operator for vectors.
    a. self – the left operand, the first vector to be added.
    b. other – the right operand, the second vector to be added.
    c. The returned value should be a new vector that is the sum of the left and right operands.
    d. Raise a ValueError exception if the vectors are not the same length.
    e. Iterate over the elements of both vectors to calculate the sum vector.
3. `__sub__(self, other)` – overloads the '-' operator for vectors.
    a. self – the left operand, the first vector to be subtracted.
    b. other – the right operand, the second vector to be subtracted.
    c. The returned value should be a new vector that is the result of subtracting the right operand from the left operand.
    d. Raise a ValueError exception if the vectors are not the same length.
    e. Iterate over the elements of both vectors to calculate the resulting vector.
4. `__mul__(self, other)` – overloads the '*' operator for multiplying vectors with vectors or vectors with a scalar.
    a. self – the left operand, the vector to be multiplied.
    b. other – the right operand, this will either be a scalar or a vector.
    c. The returned value will be:
        i. a scalar if 'other' was a vector, resulting from calculating the dot product.
        ii. a vector if 'other' was a scalar, resulting from doing scalar multiplication.
    d. Raise a ValueError if 'other' was a vector and is not the same length as the self vector.
5. `__rmul__(self, other)` – overloads the reverse '*' operator for multiplying a scalar with a vector.
    a. self – the left operand, the scalar to be multiplied.
    b. other – the right operand, the vector to be multiplied.
    c. The returned value will be a vector, resulting from doing scalar multiplication.

Notes:

1. Please place your name in the comment block at the top of the provided file.
2. Fill in the necessary code to complete the functions. Write your code where specified (ie. the TODOs). Do not modify the file name, function names, or parameters otherwise the program and tests will not work. Return the results in the format specified (ie. correct types and container types), otherwise the program and tests will not work properly.
3. Your solutions may not use any libraries or functions that perform the equivalent task.

4. Test your code using the provided ‘main.py’ file. You can write additional tests in the ‘your_tester.py’ file to further check your solutions. Note: just because the main tests pass does not guarantee that it will pass the tests on Gradescope.
5. When you are finished, submit your ‘pa7.py’ file to Gradescope on Canvas for further testing and grading. It should allow you to make repeated submissions.